ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ROBOTICS AND INDUSTRIAL AUTOMATION

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Section A: objective questions (30) · Section B: programmable logic controllers, ladder logic with timers and counters, industrial robot cells, safety in automation, SCADA, HMI and Industry 4.0 (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Ladder diagrams may use any standard addressing (e.g. I0.0 / Q0.0 or X0 / Y0), provided it is consistent.
5. Show ALL calculations.
6. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The correct order of a PLC scan cycle is …

A update outputs → read inputs → execute program.
B execute program → read inputs → update outputs.
C read inputs → update outputs → execute program.
D read inputs → execute program → update outputs.

1.1.2 In ladder logic, the symbol –| |– represents a …

A normally closed contact.
B normally open contact.
C output coil.
D timer.

1.1.3 A timer whose output switches on after a preset time once its input is on is a/an …

A off-delay timer (TOF).
B counter.
C one-shot only.
D on-delay timer (TON).

1.1.4 The stop button in a PLC motor circuit is normally wired as a …

A normally open (NO) push button.
B normally closed (NC) push button.
C toggle switch only.
D proximity sensor.

1.1.5 A PLC programming language that looks like electrical relay diagrams is …

A structured text (ST).
B assembly language.
C ladder diagram (LD).
D HTML.

1.1.6 The touch screen an operator uses to control and monitor a machine is the …

A HMI. B PLC rack. C VSD. D UPS.

1.1.7 A system that supervises and collects data from many machines or sites from a central control room is …

A CAD. B SCADA. C CNC. D ERP only.

1.1.8 A light curtain on a robot cell is a …

A lighting system for the operators. B camera for inspection.
C type of conveyor. D safety device that stops the robot when the beams are broken.

1.1.9 Connecting machines and sensors to the Internet to share data and make smart decisions is part of …

A Industry 4.0 (the Internet of Things). B the first industrial revolution.
C manual production. D hard automation only.

1.1.10 A count-up counter (CTU) with a preset of 12 switches its output on when …

A it has counted 11 pulses. B it is reset.
C it has counted 12 pulses. D 12 seconds have passed.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 A PLC can be reprogrammed without rewiring the machine.

1.2.2 A latching (seal-in) contact keeps a motor running after the start button is released.

1.2.3 Emergency stop buttons may be wired only through the PLC program.

1.2.4 Digital inputs to a PLC are usually 24 V DC in industry.

1.2.5 An off-delay timer keeps its output on for a set time after its input switches off.

1.2.6 A robot cell guard door should have an interlock switch.

1.2.7 Fixed (hard) automation is flexible and easy to change for new products.

1.2.8 Predictive maintenance uses data from sensors to forecast when parts will fail.

1.2.9 A relay output module can switch both AC and DC loads.

1.2.10 Cobots never need a risk assessment because they are always safe.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Part of the PLC that executes the program	A	CPU
1.3.2	Internal bit used to store an intermediate result in a program	B	digital twin
1.3.3	Instruction that gives one pulse when its input changes from OFF to ON	C	machine vision
1.3.4	Output of a PLC that uses a transistor, suitable for fast DC switching	D	memory bit (marker)
1.3.5	Running the program while viewing the live state of contacts and coils	E	transistor output
1.3.6	Computer-controlled machine tool, e.g. a lathe or mill	F	rising-edge (one-shot)

1.3.7	Mobile robot that moves materials around a factory automatically	G	AGV / AMR
1.3.8	Camera system that inspects products automatically	H	lock-out / tag-out
1.3.9	Isolating and locking energy sources before maintenance	I	online monitoring
1.3.10	Digital copy of a machine or factory used for simulation	J	CNC machine

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: PROGRAMMABLE LOGIC CONTROLLERS

2.1 Name FIVE main parts of a PLC system and state the function of each. (10)

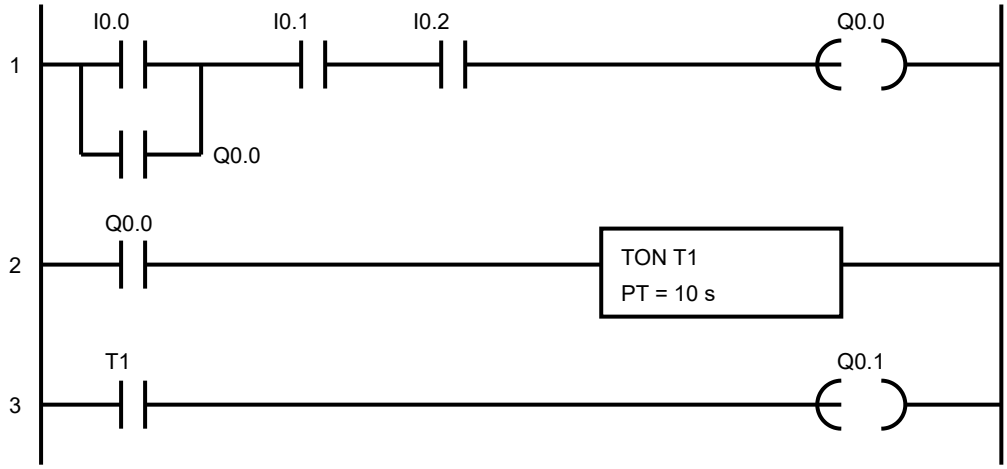
2.2 Give FOUR advantages of PLCs over hard-wired relay control. (4)

2.3 Classify each device as a PLC input or output and say whether it is digital or analogue: (a) start push button; (b) temperature transmitter (4–20 mA); (c) solenoid valve; (d) VSD speed reference (0–10 V). (4)

[18]

QUESTION 3: LADDER LOGIC

3.1 Study the ladder diagram below. I0.0 = START (NO), I0.1 = STOP (NC button, wired so the input is ON when not pressed), I0.2 = overload (input ON when healthy), Q0.0 = conveyor motor, Q0.1 = warning lamp.



3.1.1 Explain the purpose of the Q0.0 contact connected in parallel with I0.0 in rung 1. (2)

3.1.2 Describe what happens from the moment START is pressed until 10 seconds later. (4)

3.1.3 What happens to the motor and the lamp when STOP is pressed? (2)

3.1.4 Why is the STOP button wired as a normally closed button in the field? (2)

3.2 Draw a ladder diagram for the following: a filling machine uses a count-up counter (C1) to count bottles with a sensor I0.3. After 24 bottles, output Q0.2 (box full lamp) switches on. A reset button I0.4 resets the counter. (6)

3.3 Draw a ladder diagram in which a motor (Q0.3) can be started from either of two START buttons (I1.0, I1.1) and stopped from either of two STOP buttons (I1.2, I1.3, NC in the field). (6)

[22]

QUESTION 4: INDUSTRIAL ROBOT CELLS AND SAFETY

A food factory installs a robot cell in which a six-axis robot picks sealed packets from a conveyor and places them into boxes. A PLC controls the conveyor and communicates with the robot controller. Operators must sometimes enter the cell to clear jams.

4.1 Name FIVE components of the robot cell and the function of each. (5)

4.2 Describe FOUR safety measures that must be in place before operators may enter the cell. (8)

4.3 Explain how the PLC and the robot controller exchange signals (handshaking) so that the robot only picks when a packet is in position. (3)

[16]

QUESTION 5: SCADA, HMI AND INDUSTRY 4.0

5.1 Explain the difference between an HMI and a SCADA system. (4)

5.2 Name FIVE pieces of information that should be shown on the HMI screen of the robot cell. (5)

5.3 Explain THREE ways in which Industry 4.0 technologies can improve this factory. (3)

5.4 Give TWO risks of connecting factory machines to the Internet and ONE way to reduce each. (2)

[14]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER